UTAH DEPARTMENT of ENVIRONMENTAL QUALITY
AIR QUALITY

Air Monitoring Quality Assurance Program Plan

2023

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List of Abbreviations and Acronyms

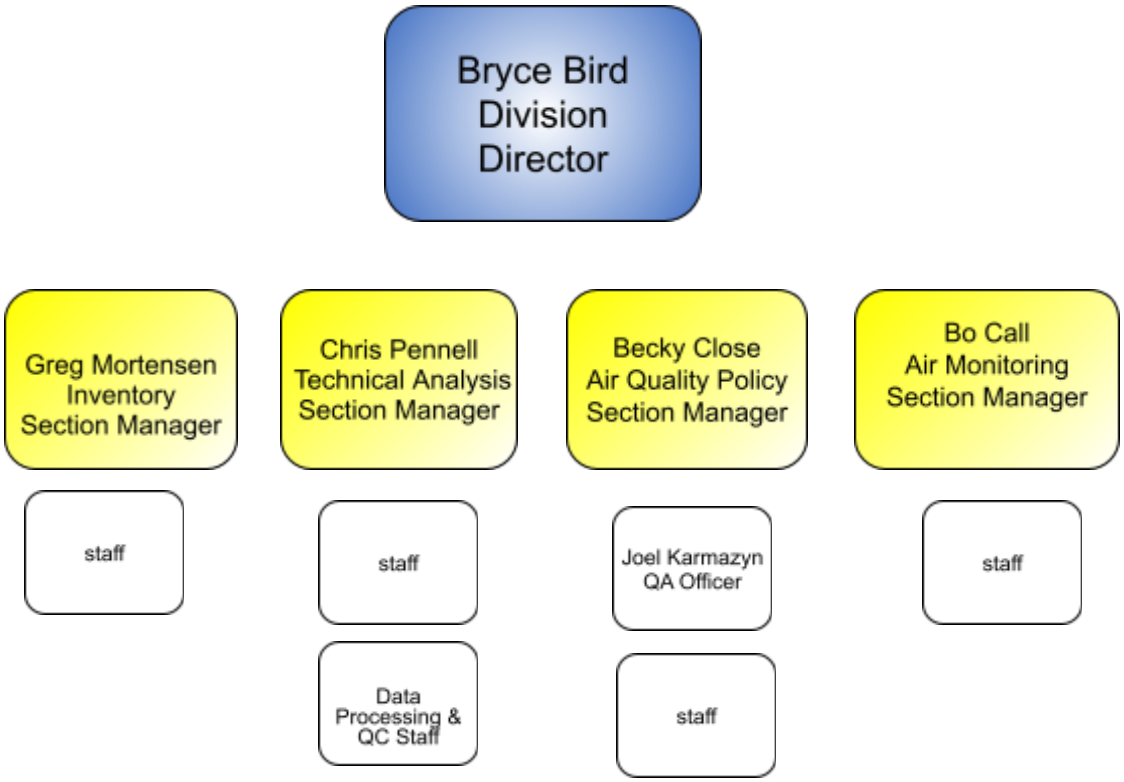
auto-GC	Automatic Gas Chromatograph
AQS	Air Quality System
AWS	Automated Weighing System
CAPs	Criteria Air Pollutants
COC	Chain of Custody
CDS	Chromatography Data System
CFR	Code of Federal Regulations
CL	Confidence Limit
CV	Coefficient of Variance
CVS	Calibration Verification Standard
DAQ	Division of Air Quality
DAS	Data Acquisition System
DEQ	Department of Environmental Quality
DNPH	2,4-dinitrophenylhydrazine
DQO	Data Quality Objectives
DVM	Digital Volt Meter
EPA	Environmental Protection Agency
FEM	Federal Equivalent Method
FID	Flame Ionization Detector
FRM	Federal Reference Method
GC	Gas Chromatograph
LCS	Laboratory Control Sample
MDL	Method Detection Limit
MQO	Measurement Quality Objectives
NAAQS	National Ambient Air Quality Standards
NADP/NTN	National Atmospheric Deposition Program/National Trends Network
NAMS	National Air Monitoring Stations
NATTN	National Air Toxics Trends Network
NIST	National Institute of Standards and Technology
NO_y	Total Reactive Nitrogen
NPAP	National Performance Audit Program
PEP	Performance Evaluation Program
QA/QC	Quality Assurance/Quality Control
QAPP	Quality Assurance Project Plan
RPD	Relative Percent Difference
SLAMS	State and Local Air Monitoring Stations
SOP	Standard Operating Procedure
SPM	Special Purpose Monitoring
SPMS	Special Purpose Monitoring Stations
SRMs	Standard Reference Materials
TNMOC	Total non-methane Organic Carbon
UHPLC	Ultra-high-performance Liquid Chromatograph
UV	Ultraviolet
VOC	Volatile Organic Compound

Utah Monitoring Network Quality Assurance Project Plan

Project/Task Organization

The Air Monitoring Section is one of ten sections in the Utah Division of Air Quality and falls under the agency's Planning Branch. The Planning Branch also includes the Inventory, Technical Analysis, and Air Quality Policy Sections. Figure 1 shows the organizational structure of the Planning Branch.

Planning Branch Organizational Chart



The Air Monitoring Section is loosely divided into the following five areas of responsibility:

- Connectivity support
- Field operations including calibrations
- Special studies
- Filter handling, weighing, and speciation
- Field audits

Background

It is the State of Utah's goal to maintain a highly reliable, ongoing ambient air monitoring network that meets the requirements of the Code of Federal Regulations, especially 40 CFR Parts 50, 53, and 58. The 1970 Clean Air Act and its amendments provide the framework for an ambient air monitoring network that is designed to collect data addressing five basic needs:

1. To activate emergency control procedures that prevent or alleviate air pollution episodes;
2. To provide air pollution data to the general public in a timely manner;
3. To judge compliance with and/or progress towards meeting ambient air quality standards;
4. To observe pollution trends throughout the region, including non-urban areas; and
5. To provide a database for research evaluation of effects: urban, land-use, transportation planning, development and evaluation of abatement strategies, and development and validation of diffusion models.

DAQ collects monitoring data for six criteria pollutants, i.e., sulfur dioxide (SO_2), carbon monoxide (CO), ozone (O_3), nitrogen dioxide (NO_2) and particulate matter (PM_{10}, and $PM_{2.5}$). In addition, DAQ operates a gas chromatograph for the collection and analysis of ozone precursor chemicals for the photochemical assessment and monitoring station (PAMS) national network. The PAMS program quality assurance plan is an appendix to the DAQ network plan.
New sites are added when there is a need determined for additional data, and retired when data requirements have been met. Each year, a network review is performed by staff and submitted as a separate document to EPA Region 8 for approval. Monitoring is done as part of the $PM_{2.5}$ Chemical Speciation Network in cooperation with EPA's Office of Air Quality Planning and Standards. Under EPA's National Air Toxics Trends Network (NATTN), DAQ monitors for gaseous volatile organic compounds, carbonyls, and semi-volatile compounds every sixth day at the Bountiful monitoring station. In addition, Utah has established a comprehensive meteorological monitoring network to supply data for modeling activities.

Project/Task Description

Air Monitoring staff are responsible for operating and maintaining an ambient air monitoring network that protects the health and welfare of the citizens of Utah. DAQ operates a network of monitoring stations along the Wasatch Front from Smithfield to Spanish Fork, in the Uinta Basin from Roosevelt to Vernal and in southern Utah in Moab, Hurricane and Enoch. In addition, special-purpose monitoring is conducted at the Price station. Monitoring stations are sited to measure air quality in both neighborhoods and industrial areas. Monitoring sites may come and go and change as needed without the need to update this document.

Ambient air monitoring is fundamental to Utah's air quality program and is used to:

1. Determine the relationship of existing pollutant concentrations to the National Ambient Air Quality Standards (NAAQS). Please refer to the Annual Network Monitoring Plan for the station locations and parameters tested;
2. Demonstrate the effectiveness of control strategies implemented to reduce and maintain criteria air pollutant concentrations below their NAAQS levels;
3. Track the effectiveness of the identified strategies and resultant regulations; and
4. Provide air pollution information for the daily Air Quality health advisories, winter season wood burning conditions, and summer season "Ozone Action Day" levels.

To be useful in assessing Utah's air quality trends and for developing pollution control strategies, collected, validated, and verified monitoring data are submitted to EPA's Air Quality System (AQS) and Air NOW databases from which reports are generated including the following:

1. Raw data reports;
2. NAMS and SLAMS reports;
3. Precision and accuracy reports;
4. Capture and missing data reports;
5. Air pollution index reports; and
6. Air quality trend reports.

Data Quality Objectives and Criteria

The primary purpose of DAQ's ambient air monitoring network is to determine whether or not primary and secondary NAAQS for criteria pollutants are being violated. Other purposes for air monitoring include, but are not limited to determining the impact of point sources on air quality, establishing background concentrations of CAPs, and determining the extent of regional ozone transport in the State.

Filters collected for the chemical speciation network program, are analyzed by EPA subcontracted laboratories and are responsible for meeting required DQOs. The Air Monitoring Section is responsible for routine maintenance, calibration, installation, and repair of the instruments.
The goal of DAQ's Air Monitoring Section is to produce data that are complete, comparable, representative, precise, and accurate. Data that meet these data quality objectives have utility in determining compliance with NAAQS. Per 40 CFR 58 Appendix A, the EPA has established Measurement Quality Objectives (MQOs) to quantify whether or not collected data adequately meet data quality objectives for each of the following five areas:

Completeness – Completeness describes the amount of valid data obtained from a measurement system compared to the amount that was expected to be obtained under correct, normal conditions. To satisfy this MQO, a minimum of 75 percent of the total possible data for the specified time period should be collected and validated. Realistically, this allows some data loss due to down time for maintenance, audits, calibration, instrument problems, inaccuracy, imprecision, operator error, etc. The completeness MQO apply to hour-, 3-hour-, 8-hour-, 24-hour-, quarter-, and annual-averaged data. Forty-five (45) minutes of valid data are required for each hour; 18 valid hours for each 24-hour sample; four days out of five per month for every sixth-day sampling; etc.

Comparability – Comparability is a measure of the confidence with which one data set or method can be compared to another. To ensure comparability, DAQ will report particulate data in micrograms per cubic meter ($\mu g/m^3$) at local conditions for $PM_{2.5}$ and at standard conditions for $PM_{10;}$ gaseous criteria pollutants in parts-per-million (ppm) or parts per billion (ppb); and flow rates in liters per minute (ℓpm). SLAMS and NAMS data will only be collected via EPA-designated federal reference or equivalent methods. Collected gaseous and particulate data will be amenable to standard statistical techniques listed in 40 CFR Part 58 Appendix A for precision and bias.
Representativeness – This term refers to the degree to which data accurately and precisely represents a characteristic of the population, a parameter variation at a sampling point, a process, or a condition. These population uncertainties can be minimized by selecting the

appropriate scale of the monitoring area consistent with monitoring objectives, by selecting suitable sampling schedules and frequencies, and by identifying physical constraints imposed by meteorology, topography, emission sources, and land access.

Precision – Measure of agreement among repeated measurements of the same property under identical, or substantially similar, conditions. The DQO for precision for gaseous parameters is set forth in EPA's Quality Assurance Handbook for Air Pollution Measurement Systems, Volume II, Appendix D, 2013, at 90%. This term refers to a confidence limit of the coefficient of variation (90% CL CV). For CO and SO_2 the precision estimate (90% CL CV) should be ≤ 10%; for ozone ≤ 7%; and for NO_2 ≤15%. These calibrations and precision checks will be made and calculated semi-annually. For collocated PM_{10} and $PM_{2.5}$ monitors, the precision estimate, CV, should be equal to or less than 10% with annual and three-year estimates at 90% CL of CV ≤ 10%. Collocated PM monitors should be operated at a minimum 15% of available PM sampling sites and run at least every 12th day for PM_{10} and $PM_{2.5}$.

Bias – Bias is the systematic or persistent distortion of a measurement process that causes error in one direction. It is estimated from the positive and negative deviation from the true value as a percentage of the true value. To minimize systematic operator error, it may be valuable to rotate field operators to different sampling runs on a regular basis. Systematic error may be introduced if field operators remain assigned to the same run for prolonged periods of time, if the same flow transfer device is always used in flow verification audits, or if insufficient time lapses before a stable concentration can be read when audits or calibrations are performed. A one-point precision check is made at least weekly of each operating gaseous monitor against a bottle of NIST-traceable gas of known concentration or in the case of ozone against a transfer standard. The absolute bias estimate of these single point QC checks at a 95% confidence level should be ≤ ±7% for ozone, ≤ ±10% for CO, and SO_2, and ≤ ±15% for NO_2. Performance Evaluation Program audits performed quarterly by EPA contractors should not be greater than ±10% for $PM_{2.5}$. If the relative percent differences are rank-ordered with their "±" differences, then the 25th and 75th percentiles should have different signs. If both percentiles are negative or positive, this is strong indication that the weekly or bi-weekly checks show bias and the problem must be investigated by the data quality analysts and as necessary, by the QA Officer.

Data quality will be calculated at least annually according to EPA's accepted statistical procedures to determine compliance with the recommended limits. Data outside these limits will still be reported to AQS but DAQ will flag the data internally and attempt to determine the source of the problem(s). Data outside these limits may still be used for qualitative analysis purposes based on the "weight of evidence" of data suitability as per 40 CFR 58, Appendix A, 1.1.2.3.
Monitoring data are used for the following activities:

- Attainment/non-attainment demonstration
- Air quality trends
- Air quality modeling
- Air Pollution Index reporting
- Reports to the public

Users of DAQ's monitoring data include:

- Federal, state, and local agencies
- Industry
- Environmental groups
- General public
- Research groups
- Media

General Training Requirements

The personnel involved in air monitoring and data analysis functions should have sufficient training in their jobs to perform their functions at an acceptable level. The necessary training will be evaluated in terms of employee need and functional level required. Training will be provided to remote station operators and new employees. Training of existing staff will be provided as needed or upon introduction of new instrumentation, software, methods, or procedures. In addition to in-house training by DAQ staff and equipment vendors, staff will be encouraged to attend seminars, courses, and workshops provided by EPA and its contractors.

New Monitoring Section Personnel Training Procedures: New Monitoring Section personnel will receive instructions on Department, Division, and Monitoring Section policies. New personnel will become familiar with the objectives of the monitoring program and how it is structured, read and understand the regulations, guidelines, and procedures applicable to ambient monitoring; instructed in the use of materials and tools needed to operate the program; trained in fieldwork, data acquisition systems, data reduction, verification and validation, and data quality assessment; learn general laboratory procedures and how to interact with other agencies, industry, and the public, and receive safety instruction in all areas where potential danger exists.

Training in the Maintenance, Troubleshooting and Repair of Ambient Air Monitoring Equipment: Upon completion of training, each trainee will be able to:

- Understand the function and operational theory of monitoring equipment
- Perform each required maintenance task in the maintenance schedule
- Properly use troubleshooting guides and equipment manuals
- Identify symptoms of problems as they arise
- Troubleshoot and identify the source of problems
- Repair/correct problems.

Basic Knowledge and Skills Needed QA/QC:

- Be educated in scientific principles
- Understand, in detail, all tasks required to operate ambient air monitoring systems
- Be familiar with air monitoring technology
- Fully understand QA/QC principles and guidelines contained in 40 CFR Part 58 and apply them to each task performed
- Be proficient in identifying problems and deficiencies in the operation of the monitoring program and provide solutions
- Be able to write QA/QC procedures for new sampling and data reduction methods consistent with federal regulations and guidelines
- Possess a basic understanding of statistical methods and computational skills
- Be able to develop ambient study protocols on an as needed basis.

Field Operations:

- Be familiar with basic principles of instrument operation, trouble- shooting and repair
- Be able to follow standard operational procedures and perform routine QC tasks
- Be able to identify instrument problems in the field and perform on-site repairs
- Be able to train remote operators to perform routine checks on instrument operation and routine checks on station status
- Be proficient in site installation
- Be able to perform preventative maintenance according to established schedules and procedures.

Data Processing:

- Have knowledge of data reduction methods
- Be able to identify invalid data based on data reduction procedures
- Fully understand instrument operations and electronic data signal processing
- Be familiar with telemetry systems
- Be able to input data into database systems
- Be able to generate reports and summary statistics from database systems
- Be familiar with data chain-of-custody procedures and forms
- Be able to maintain and update site and monitoring information in AQS when data changes.

Particulate Filters:

- Be able to process filters according to established SOP
- Be familiar with the principle of particulate sampler operations
- Be able to program sequential samplers with new batch of tared filters and to download data from monitor following batch run
- Be able to fill out sample cards with relevant information and return filters to the freezer at the Technical Center to preserve captured volatiles
- Be able to troubleshoot equipment and take corrective action
- Be able to perform monthly flow verifications, flow audits and yearly calibrations of both manual and automated, filter-based and continuous analyzers.

Documentation and Records

Site Files: As part of the site documentation, each monitoring site will have a site file. Maintenance of site files is the responsibility of a designated scientist assigned to the Monitoring Section. Site files are located at the Technical Center and contain the following information:

1. Reason for establishing the site, method used to determine site location, e.g., modeling, traffic counts, saturation study with accuracy estimate.
2. A Site Information document which contains the following:
   - The parameters measured
   - Exact location of the site, including UTM coordinates, site elevation and description of the shelter
   - Description of environment around site
   - Point and area sources in the vicinity
   - Probe placement

- Heights and distances of obstructions
- Topography of the area
- Monitoring objective and spatial scale
- NCore, SLAMS, or SPM site
- Photographs of the site and from the site in each of the four cardinal directions
- A detailed map of the area
- Wind rose, if available
- Start up and shut down dates for each parameter or changes in operation for any parameter
- Property leases or agreements and local permits where required
- Station operator contracts
- Any other information pertinent to site or station, such as names and phone numbers of landowners, lessors, or neighbors, station operator information, special site information, power and internet arrangements

Major tasks performed in the monitoring program and other significant tasks deemed appropriate are documented and the records are filed at DAQ under the general supervision of the Air Monitoring Section manager.

QA/QC Records: Quality control records ─ particulate monthly flow verification audits, weekly precision/zero/span checks of gas analyzers, and calibrations ─ are directly reviewed, maintained, and filed by staff assigned to data processing. Validation records for missing data, data outliers, and data flags for maintenance and unusual events are maintained and filed by data processing staff as they review incoming data. Data is backed up daily.

Filter Records: Electronic data from the continuous particulate monitors are downloaded directly by the data loggers into AirVision. The electronic data from the filter based particulate monitors are downloaded weekly or biweekly (depending on site location) by field operators on flash drives. These records include filter identification number and instrument operating parameters such as total sample time, average ambient pressure and temperature, and total sample volumes at local and standard conditions. After checking these field-generated data for validity, the laboratory operator uploads the volumetric data into DAQ's automated weighing system (AWS). Actual and standard concentrations are calculated by the AWS in a summary report. Data is backed up daily. Gravimetric and volumetric data are verified by the data processing staff before being entered into AQS.

Air Monitoring Record Maintenance and Retention Policy: The following air monitoring record maintenance and retention policy list enumerates the documentation and records which must be maintained and retained by the Air Monitoring Center for data to be scientifically correct and legally defensible:

Retention Policy

Documentation	Retention in Years
Quality Assurance: Certifications: NIST, In-house, Testing Labs, Manufacturer, Calibrator Side by Side. Audits: System, Performance, Precision. QUAP	 5 5 Current
Quality Control Equipment Calibration, Data Flags, Station Status Reports, Instrument Span Checks, Maintenance Records, Log Books, PM Data Cards.	5 (plus life of instruments)
Data Weekly/monthly hourly Lab Reports, Data Capture.	 5 Permanent
Equipment Specifications, Manuals, etc.	Life of Equipment
PM Filters Exposed PM2.5/PM10 PM2.5 That Exceed NAAQS	 2 Years Cold Storage, then 1 Year Onsite Storage Followed by 7 Years Archiving Permanent, except When Used for Special Studies
Station Files	7 Years After Closing

Personnel Certification Records: All equipment operation certification records required by personnel job functions are to be kept by the employee and a copy provided to the Branch Secretary for filing and to their supervisor.

Schedule of Deliverables to EPA

Ambient Data	60 days after each calendar quarter
Precision and Accuracy	90 days after each calendar quarter
SLAMS Report	July 1, of each year
Revisions to the QAPP	every 5 years or as required by major changes
Annual Network Review	July 1, of each year

All Ambient Data, Precision and Accuracy, and SLAMS deliverables are submitted through the AQS. Revisions to the QAPP and Annual Network Plan are submitted to EPA Region 8 Air Program

Sampling Process Design

Selection and operation of monitoring sites are critical in assuring that the data collected are representative and comparable. All selected sites operated by DAQ must meet appropriate federal requirements listed in 40 CFR Part 58. In the event that a monitoring site no longer meets siting criteria, DAQ will consult with EPA on the appropriate course of action. DAQ employs

information and procedures outlined in EPA's guidance documents for site selection and operation of air quality and meteorological parameters. The following requirements should be considered in designing the sampling process:

- Network design for establishing NCore, SLAMS, and SPMS sites (40 CFR Part 58, Appendix D)
- Probe siting criteria (40 CFR Part 58, Appendix E)
- Quality assurance requirements (40 CFR Part 58 Appendix A)
- Ambient air quality monitoring methodologies (40 CFR Part 58, Appendix C)
- Reference methods for ambient air quality sampling (40 CFR Part 50, Appendices A, B, C, D, F, G, J, L, and Q)
- Ambient monitoring equipment selection and evaluation (40 CFR Part 58.11, 58.12 and Appendix A)
- Data acquisition system (40 CFR Part 58.16, Appendices A and E, data submittal to AQS)

The purpose for monitoring and why specific locations were selected is explained in DAQ's Annual Network Plan per 40 CFR Part 58 Appendix D. The Plan is posted on the Agency website.
Siting network monitoring stations draws upon such inputs as emissions inventory data, results from climatologic dispersion modeling, population and population changes, growth of industrial and area sources, traffic patterns, determining the monitoring objective, and identifying the appropriate spatial scale required to meet the monitoring objective. Input from the Inventory Section, Technical Analysis Section, Policy Section, and Air Monitoring Sections are required in identifying the general and specific locations where a sampling site should be placed. The sampling process design must comply with all requirements contained in:

1. 40 CFR Part 58, Appendix A (*Quality Assurance Requirements for SLAMs, SPMs, and PSD Air Monitoring Program*)
2. 40 CFR Part 58, Appendix C (*Ambient Air Quality Monitoring Methodology)*
3. 40 CFR Part 58, Appendix D (*Network Design Criteria for Ambient Air Quality Monitoring*)
4. 40 CFR Part 58, Appendix E (*Probe and Monitoring Path Siting Criteria for Ambient Air Quality Monitoring)*
5. 40 CFR Part 50, Appendices A, B, C, D, F, G, J, L, and Q (Reference Methods)
6. *Quality Assurance Handbook for Air Pollution Measurement Systems, Volume IV, Meteorological Measurements*, Version 2.0, March 2008.
7. *On-site Meteorological Program Guidance for Regulatory Modeling Applications*, EPA-450/4-87-013, June 1987

Detailed station installation procedures are contained in Section A.3.0 of DAQ's Utah Ambient Air Monitoring Procedure Manual.

<u>Instrument Selection and Acceptance</u>: Air sampling instruments are highly sophisticated, measurement devices designed to quantify pollutant concentrations in ambient air at trace levels. Great care must be taken prior to and after purchasing an instrument to test its reliability and capability to collect accurate and precise monitoring data.

Utah Division of Air Quality Air Monitoring: Staff are uniquely positioned to understand the challenges of operating a network that traverses dynamic geographic (including dramatic elevation changes), climatic and complex terrain that directly impact instrument performance. The aforementioned environmental factors impact air monitoring instruments in accordance with the

laws of physics. That is to say that an instrument in the Wasatch Front at an elevation of 4,000 feet will operate dramatically different than the same instrument located in Escalante at an elevation of 6,000-6,500 feet. Likewise, the same instrument will operate differently when placed in Cache Valley during the winter months because of the propensity for fog in that valley. Therefore, it is imperative that prior to adding new instruments into the air monitoring network that the selection criteria must go beyond the minimum requirements specified in 40 CFR Part 53. Specifically, we must have an opportunity to test instruments across all of the expected conditions that we experience in Utah before making the decision to include an instrument into the network. This will require the purchase of representative instruments to test in our unique environmental conditions and within our operation parameters.

Utah's air monitoring network is nearly spread across the 350 miles long and 270 miles wide state. Instrument reliability is an import factor when making a purchase selection as it reduces the amount of service required at a monitoring station. The ultimate cost of an instrument goes far beyond the capital cost when considering the labor and travel time required to maintain instruments that have a poor reliability track record. Unreliable instruments are not cost-effective therefore; we must be able to assess instrument long term reliability in the field before purchase.
Instrument database networking options, inter-phasing, internal data storage capability and internal diagnostic options and ease of use are additional cost-effectiveness factors that should be considered before inclusion into the network as they have extensive after purchase consequences.
It is imperative that we stay abreast of emerging technology in order to maintain a state-of-the-art network. While we encourage new development, we must be assured that the technology will operate as designed in our complex system. New technological advancements should be thoroughly tested in our environment before making procurement decisions.
Instruments supplied by vendors for evaluation or procured for evaluation are first set up in the Technical Center (AMC) instrument laboratory and evaluated for the following characteristics:

1. Accuracy
2. Precision
3. Reliability
4. Noise
5. Quality of assembly
6. Ease of access to components
7. Compatibility with DAQ's telemetry system and other support equipment
8. Manufacturer specification, where possible
9. Ease of use.

Upon completion of the evaluation, instruments are ranked according to the test results, manufacturer repair support, and spare parts availability. Subsequently, instruments that are deemed fit to use at the AMC are then moved into the field for testing under the rigors of the Utah network. Instruments that do not appear to meet the base requirements or that fail will not be field tested further. However, we will continue to work with willing vendors to remedy any deficiencies discovered during testing. While final cost is always considered, it is of minimal importance when instruments are selected for inclusion in the monitoring network.

An instrument's ongoing acceptability is based on its performance as determined from calibrations, audits, span checks, breakdown records, and repair intervals. A log is chronologically maintained

for each purchased instrument to document replacement of parts, major maintenance performed, and audit/calibration performance.

Sampling Methods

All standard operating procedures for the collection of ambient data are contained in the Utah Ambient Air Monitoring Procedure Manual B and C.

Sampling Handling and Custody

Sample Handling: Sample handling relates to filter-based samples ($PM_{2.5}$, PM_{10}), semi-volatile organic compounds, the collection of VOCs in Summa canisters, and $PM_{2.5}$ samples collected on Teflon filters in canisters to be speciated. Tared and exposed filters are stored in pre-cleaned Petri dishes/slides. Laboratory-supplied containers must be properly labeled. DAQ will purchase only pre-numbered and bar-coded Teflon filters from manufacturers. All collected samples must be carefully removed from the monitor, the accompanying paperwork completed, and when required, a tamper-proof custody seal affixed. PM filters are transported back to the Technical Center in coolers by staff. To avoid contamination and sample invalidation, unexposed and exposed Teflon for $PM_{2.5}$ and PM_{10} collection must never be touched. They should only be handled with Teflon-coated forceps.

Holding Times: DAQ will make every effort to comply with maximum filter holding times per QA Handbook recommendations, Volume II, Appendix D, however, any filters that exceed recommended holding times will be evaluated on a weight of evidence approach to determine the validity of the data.

Holding Times

Pollutant	Criterion	Acceptable Range
$PM_{2.5}$, PM_{10}	Sample Recovery	≤ 7 days 9 hours from sample end date.
$PM_{2.5}$ PM_{10}	Post-sampling Weighing	≤ 10 days from sample end date if shipped at ambient temperature or ≤ 30 days if shipped at below ambient or ≤ 4°C or below for average sampling temperatures < 4°C from sample end date. PM_{10} is not limited to 30 days. Post-sampling should be performed as soon as possible.
$PM_{2.5}$, PM_{10}	Filter Equilibration	24 hours Minimum
$PM_{2.5}$, PM_{10}	Pre-sampling Filters	≤ 30 days before Sampling
$PM_{2.5}$, PM_{10}	Pre-analysis Storage	Indefinite if stored at ≤ 4°C, or ≤ 30 days if stored at ambient temperature.

Chain-of-Custody: Data chain-of-custody is defined as a process by which standards, samples, data, and documentation are transferred and stored securely by air monitoring personnel. The intent for establishing a data chain-of-custody is to collect legally defensible data useful for regulating ambient air quality. Signatures for transferring material between staff members are not required since the Monitoring Section operates as a homogenous group. However, all quality assurance and quality control documents are initialed and dated by the person entering information. Signatures for transferring material to and from other agencies and laboratories are required and have been implemented according to chain-of-custody procedures established by EPA (see Quality Assurance Handbook for Air Pollution Measurement Systems Vol. II, Section 8.2, December 2008).

The following chain-of-custody criteria have been established:

1. Each staff member has custody of standards, samples, data, and documentation for the time period staff uses them to perform an operational task.
2. All quality assurance and quality control documentation must be dated and initialed at the time an entry is made.
3. Samples for laboratory analysis, including sample documentation, must be hand-carried to the laboratory or shipped as a sealed package.
4. Results from analyzed samples must be picked up from the laboratory or sent by them in a sealed envelope. The recipient must sign and date the chain-of-custody form
5. All monitoring stations must be secured by fences where possible and remain locked at all times when not occupied by monitoring staff. Keys are only issued to authorized personnel.
6. Samples, data, and documentation must be secured at the Technical Center/headquarters before and after analysis per the Division retention schedule.
7. Calibration/Audit standards and their documentation establishing traceability to NIST reference material must be kept in a secure place at the Technical Center. Access is only permitted to monitoring/auditing staff.
8. Implementation and maintenance of data chain-of custody procedures are the responsibility of the Monitoring Section manager.

Analytical Methods

All of the analyzers and sampling methods for the criteria pollutants used by the DAQ are reference or equivalent methods.

Quality Control

Quality control (QC) is a systematic approach to maintain desired standards for the collection, validation, and verification of ambient pollution data. This requires QC activities at regular intervals for data collection, data management, and reference materials.

The frequency of QC checks is determined by how often QC limits are exceeded. If desired standards for data accuracy/bias, precision, sensitivity, and data capture can be maintained, corrective measures are required less often. The minimum QC checks are in accordance with those specified in 40 CFR, Part 58, Appendix A and guidance established in the QA Handbook Volume IV for meteorological parameters. More frequent QC checks may be performed based on the experience of the operator.

Quality Control Statistics

The following table provides the data quality indicators calculated for each measured pollutant. As part of its quality assurance program, the QA Officer will use the AQS standard report AMP 256 to evaluate the statistics outlined in 40 CFR Part 58 Appendix A, as well as other appropriate AQS reports. Standard QC reporting options available in AirVision and MTL will also be used.

Calculated Data Quality Indicators

Statistic	O_3	SO_2	NO_2	CO	$PM_{2.5}$	PM_{10}	$PM_{10-2.5}$
Precision Estimate	x	x	x	x	x	x	x
Bias Estimate					x		
Absolute Bias Estimate	x	x	x	x	x		x
Semi-Annual Flow Rate					x	x	x
One-Point Flow Rate					x	x	x

Instrument Maintenance

See weekly through annual preventative maintenance schedules listed in monitoring section's procedures manual, section B under each pollutant and meteorological parameter heading.

1. Spare Parts Replacement Schedule: Instrument components are replaced as needed (contingent upon available funding). DAQ maintains a partial spare part inventory for each make of gaseous and particulate instrument deployed in its network. Designated monitoring personnel are assigned to check these inventories on a regular basis and reorder spare parts found in short supply.
2. Inspection of New Equipment: A purchased monitor is set up and tested in the Technical Center laboratory to assure that it is complete and capable of operating according to manufacturer's specification. Testing is done according to SOP for existing instruments. For new instruments the SOP is developed after check in and instrument evaluation.

Instrument/Equipment Calibration and Frequency

DAQ will calibrate or recalibrate particulate and gaseous analyzers upon initial installation, following physical relocation, after any repairs or service that might affect its calibration, following an interruption of operation, upon any indication of analyzer malfunction or change in calibration, and at designated, regular intervals.

Calibration Frequency

Parameter	Frequency	Procedure (SOP Manual)
PM_{10}	Annual Multi-Point calibration and if out-of-control or after major repair	Section B.8.4.6
$PM_{2.5}$	Annual Multi-Point calibration and if out-of-control or after major repair	Sections B.8.7.6, B.8.8.6
$PM_{2.5}$ (URG3000N)	Annual Multi-Point calibration and if out-of-control or after major repair	Per manufacturer spec.
$PM_{2.5}$ (SASS)	Annual multi-Point calibration and if out-of-control or after major repair	Per manufacturer spec.
Wind Speed	Annual per the SOP	Met Instrument Calibration SOP
Wind Direction	NA	Met Instrument Calibration SOP
Temperature		Section B.15.2.1
Relative Humidity	Annually per the SOP	Section B.15.3.1
SO_2	Semi-annually or if out-of-control or after major repair.	Sections B.9.3.7, B.9.4.7
CO		Sections B.10.1.7, B.10.2.7
NO_2		Sections B.11.1.7, B.11.2.7, B.11.3.7
O_3		Sections B.12.1.7, B.12.2.7

Inspection/Acceptance for Supplies and Consumables

DAQ maintains an inventory of replaceable spare parts, consumables, and maintenance items. Spare parts are generally purchased directly from instrument manufacturers.

Particulate sampling filters for criteria pollutants (PM_{10}, $PM_{2.5}$ and $PM_{10-2.5}$) are purchased from EPA contractors and are pre-numbered and barcoded. Prior to weighing each filter, it is visually inspected for pinholes, tears, or other flaws. Filters with defects are discarded.

Non-Direct Measurements

Collected gaseous and particulate monitoring data are based on direct measurements. This data is input into AQS. Non-direct measurements published in scientific articles, other EPA databases, and air studies done by other groups locally and regionally are frequently used by DAQ. Examples of non-direct measurements include special studies of air pollutants published by local universities, EPA's National Emissions Inventories and Toxics Release Inventories, and speciated mercury concentrations measured by other states who submit monitoring data to the National Atmospheric Deposition Program.

To assess the quality of data provided by other sources, DAQ tries to: 1) identify the source's objectives for gathering the data; 2) assess the reliability of data sources and calculations; and 3) determine, if possible, what QA/QC measures were applied in validating and verifying data including flagging and discarding data. As a frequent user of secondary data, DAQ must establish

acceptance/rejection criteria and perform more than a cursory quality assessment of these data. Whenever possible, DAQ must discuss in its review any limitations on the use of the data (geographical limitations, different sampling or analytical methods used, availability or lack of QA/QC records). It is not sufficient to assume that since a project was performed by a reliable source, such as a peer-reviewed publication from a local university or data taken from a national database that the secondary data may be used without reservation.

Data Management

DAQ's data acquisition system for monitoring data is described in the monitoring section's SOP B14.1.2. Each monitoring site that collects air quality data is equipped with an 8816 and/or 8832 ESC data logger, modem, and desktop computer. The ESC 8800 series data logger provides a microprocessor-based data acquisition system designed specifically to acquire, process, store environmental data. Control of daily zero/span are functions that may be performed by the data logger. Acquired data is transmitted hourly via DSL or wireless internet to a server located at headquarters through AirVision software.

The Campbell Scientific CR6 loggers acquire, process, and store meteorological parameters from 2D Sonic wind and other sensors. Hourly, 15-minute and 1-minute values of wind speed, wind direction, barometric pressure, temperature, solar radiation and others are collected. The acquired data are transmitted hourly via DSL or wireless internet to a server located at headquarters and imported into AirVision software.

The AirVision software is a relational database system. It contains a server at a secured DEQ site with Internet and modem connections. The AirVision server pulls all monitoring network's data hourly resulting in a near-real time data collection and observation.

The individual data loggers provide up to 30 days backup of stored data depending on the number of air quality and meteorological parameters monitored. The AMS utilizes a second data collecting system: Digitrend and LoggerNet. The Digitrend data logger collects minute and hourly data on gaseous or particulate parameters measured at that site. Gaseous and real-time particulate data are also stored at the station computer. Digitrend data are downloaded securely onto the Technical Center server in an open directory accessible to all monitoring personnel by login and password for review and interpretation. Data is stored in a Microsoft SQL database on the headquarters server by the Department of Technology Services at the multi-agency state office building. An AQS format file is created from the central database and written to the headquarters' fileserver. Emails or data logger central messages are sent to the data coordinator when problems are found (power outage, exceptional events, etc.).

These data files are backed up daily on a local server as well as the large state server maintained by the Information Technology Group at the state capitol.

Data validation and editing is a multi-layered quality control process. Quality assurance staff assigned to data operations are responsible for reviewing and validating the monthly ESC files for each air quality and meteorological parameter measured. A separate directory on the fileserver is used for this data review. Staff review and compare each hour of air quality data from continuous gaseous and particulate analyzers as well as meteorological data collected by AirVision.

An AQS format file is created from the central database and written to the Technical Center fileserver. Validated data is entered into AQS.

Data management is primarily the responsibility of the DAS coordinators for the monitoring section. They maintain both the DAS software and hardware. They also install and test it at the various network sites, troubleshoot and fix problems as they arise, and back up the data to the server daily. Purchased hardware must be compatible with AirVision software.

Near real-time monitoring data (not quality assured) are used to: 1) make daily forecasts of criteria air pollutant levels (CAPs) in monitored counties throughout Utah; 2) forecast national daily pollution levels of CAPs in EPA's AIRNOW program; 3) control the emissions of the pollutant precursors along the Wasatch Front during wintertime inversions for $PM_{2.5}$ and summertime for ozone in DAQ's mandatory wood-burning program and "Choose the Clean Air" programs; and 4) provide data to the UtahAir mobile App.

During each field visit, field operations staff and remote operator's record critical operating parameters in the log (flow rates, zero readings, operating status of the DAS, etc.) Memos are filled out when problems are found (construction, exceptional event, etc.)

Assessments and Response Actions

The Monitoring Section can be considered a primary quality assurance organization where measurement uncertainty among all the monitoring stations can be considered reasonably homogeneous because of common factors. Common teams of field operators, calibrators, data processors, use of standard operating procedures (SOPs), common calibration facilities and standards, a common quality assurance oversight organization, and the same management, laboratory, and DEQ support. Quality assurance also addresses the assessment of the quality of monitoring data by providing statistical estimates (data quality indicators) of precision, bias, and completeness.
Quality assurance is accomplished by challenging the quality control program at regular intervals with independently verifiable quality assurance checks and by subjecting measured data to rigorous statistical analysis. If data analysis shows that data quality is inadequate, the quality control effort must be increased until data quality is acceptable. A quality assurance program includes the following areas:

1. <u>Quality Assurance Schedule</u>
   a. Weekly:
      - Perform precision checks with collocated particulate sampler.
      - Perform zero, precision and span checks on continuous gaseous analyzers.
      - Perform a visual check of the sonic sensors and sensors transducers for physical damage or changes in the orientation or and alignment.
   b. Monthly:
      - Perform a flow rate verification on each continuous and manual $PM_{2.5}$ and PM_{10} sampler.
      - Perform an external and internal leak check on $PM_{2.5}$ and PM_{10} samplers.
      - Perform a temperature and pressure check on $PM_{2.5}$ and PM_{10} samplers.
        <u>Note</u>: $PM_{2.5}$ URG3000N and SASS samplers are included here.
   c. Semi-annually:
      - Perform multi-point performance audits on continuous gaseous analyzers.

- Perform side-by-side field calibration systems accuracy tests against a primary laboratory calibration system.

d. Semi-Annually:
   - Certify analytical balance with a NIST traceable standard.
   - Certify Class 1 and Class 0 weights with a NIST traceable standard.
   - Certify Hi-vol field calibration devices with an Alicat.
   - Certify calibration gases to NIST certified standard reference materials (SRMs).
   - Perform flow audits on all FRM particulate samplers and continuous particulate monitors. These audits are assigned to two data quality control personnel.

e. Annually (external):
   - Perform $PM_{2.5}$ Performance Evaluation Program (PEP) audits at DAQ's $PM_{2.5}$ sites. External and internal leak checks and temperature and pressure sensor audits are performed as part of the PEP audit. All FRM or FEM $PM_{2.5}$ samplers must be subject to a PEP audit at least once every six years. An EPA contractor visits the sites with DAQ staff to conduct audits.
   - Certify flow rate transfer standards to a NIST traceable standard. FRMs are sent back to vendors where they were purchased for recertification.
   - Certify ozone transfer standard against a NIST traceable Standard Reference Photometer at the regional EPA laboratory.
   - Certify calibration gases to NIST certified SRMs.
   - Internal auditors conduct meteorological performance audits.
   - Perform configuration audits on monitoring network.

f. Two Years:
   - Recertification of all Standard Reference Materials (SRMs) against NIST traceable standards.

2. <u>Traceability of Audit Standards</u>

   Audit standards for gaseous and particulate parameters are either NIST certified or NIST traceable. NIST certified standards are directly purchased from NIST while NIST traceable gas standards are certified at the Technical Center laboratory against NIST SRMs or other NIST traceable primary standards. Frequency of certification depends on the stability of the standards. Ozone transfer standards are certified quarterly. Particulate audit flow devices are certified annually by the manufacturer.

3. <u>Systems Audits</u>

   At least annually, the monitoring section manager or designee visits all monitoring sites in the network to assess network compliance with federal regulations and guidelines. Each site is inspected to verify that all monitoring systems are properly sited and in good working order. DAQ uses a checklist to assist in the review.

   The audit helps identify problems such as inlet probe obstructions, lack of instrument maintenance, faulty site temperature control, dirty manifold and inlet probes, safety hazards, security risks, shelter weather leaks, missing equipment parts, frayed power cords, and checking that monitor exhausts are not being sampled. All documentation pertaining to quality control and quality assurance is inspected. If the site review indicates any problems, the section manager must immediately initiate corrective actions with the staff.

4. <u>Performance Audits Frequency and Acceptable Limits</u>

   Performance audits are performed for the various gaseous and particulate pollutants according to the following schedule by EPA contractors.

5. Configuration Audit
   Monitoring sites are audited annually in a configuration audit in which the layout of the shelter and instrumentation is evaluated for continuity with other stations in the network.

6. Technical Systems Audits
   A system audit of each monitoring organization within EPA Region 8 is conducted every three years by Regional QA staff. The on-site review assesses DAQ's air monitoring program for compliance with established regulations governing the collection, analysis, validation, and reporting of ambient air quality data. The systems audit consists of three phases: 1) pre-audit activities that describe the proposed audit plan; 2) on-site audit activities (initial meeting with section manager, interviewing key personnel, site visits, data quality audits, summarize findings, and discuss findings with key personnel and QA Officer; and 3) post-audit activities include issuance of systems audit report within 30 calendar days of audit completion and corrective action and DAQ follow-up response. The EPA Project Officer, EPA Air Monitoring Section Chief, EPA Air Monitoring State Coordinator, or the EPA Regional Quality Assurance Officer may at any time conduct an announced or unannounced audit of the environmental data collection activities conducted by or on behalf of the State. This includes audits of any contractors or subcontractors.

7. Gaseous Analyzers
   All gaseous analyzers are audited annually using certified audit systems employing NIST or NIST traceable standards by members of the audit team. NO_2 is generated by gas-phase titration with excess nitric oxide (NO). Ozone is generated with a certified UV photometer. The evaluation is made by challenging the analyzer with an audit gas standard of known concentration from at least three audit levels and zero, in accordance with 40 CFR 58 Appendix A Section 3.1.2.1. One point must be within two to three times the method detection limit of the instruments; however, EPA recognizes the difficulty of attaining this level. Consequently, EPA has advised states to do the best they can to achieve this requirement.
   The audit concentrations plus zero are introduced at the beginning of the sample train, including sample line and inlet filter, beginning with the highest concentration descending in order to the lowest concentration including zero. Each point is run for 15 to 20 minutes or until stable. The resulting indicated values plus all other pertinent information are recorded on the audit/calibration sheet. The indicated concentrations are compared to the audit concentrations by linear regression and a full-scale percent deviation value is determined.
8. $PM_{2.5}$ Samplers (includes URG3000N and SASS)
   a. Temperature: Ambient and filter temperature probes are compared at ambient temperature to an NIST traceable thermometer. If the deviation is greater than ±2°C of the standard, the temperature sensor(s) must be recalibrated.
   b. Ambient Pressure: Ambient pressure is audited with a calibrated pressure sensor at ambient conditions. This is done by collocating the audit device with the station pressure sensor. If the deviation is greater than ± 10 mm Hg of the standard, the pressure sensor must be recalibrated.
   c. Flow Verification-Audit/Actual: Flow rate verifications are performed monthly with a traceable flow transfer standard. The one-point verification is performed at the sampler's set-point flow rate with the sampler being in its normal operating configuration including the sample filter and electronic flow sensor. The flow sensor

determined set-point flow of 16.7 ℓ/min at ambient conditions is compared to the transfer standard. If the error is > ± 4.0%, then the sampler must be recalibrated.
Note: The set-point flow rate for the $PM_{2.5}$ Met One ambient chemical speciation sampler (SASS) is 6.7 ℓ/min. If the flow deviation exceeds ± 10%, then troubleshoot the problem to determine the cause of the change. Similarly, the URG3000N $PM_{2.5}$ sampler set point flow rate is 22.0 ℓ/min with a ± 10% flow tolerance. If flow uncertainties exceed ± 10%, a flow re-calibration is required. As part of the flow verification, monthly temperature, pressure, and leak check verifications are required.

d. Performance Evaluation Program Audits: The audit is performed by an EPA contractor who collocates his FRM $PM_{2.5}$ sampler next to a DAQ $PM_{2.5}$ monitor. Both samplers run at their pre-determined set point flow rates during the same
24-hour midnight-to-midnight period. Filters from both samplers must be subjected to the same field and laboratory conditions as specified in the $PM_{2.5}$ reference method. If the measured percent differences result in an absolute bias estimate
> 10%, all calibrations must be checked and the filter handling process must be reevaluated.

9. PM_{10} Samplers
A one-point flow audit must be conducted semi-annually on each PM_{10} monitor in the network. A one-point flow audit is performed with the sampler in its normal operating configuration and the particulate filter in place. The ambient set point flow rate displayed by the sampler is compared to the audit flow rate measured with a certified flow measurement device. The percent differences between the NIST traceable audit device and the measured flow rate must be within ± 4% or the analyzer must be re-calibrated. These flow rate uncertainties are used to estimate bias that should be ≤ 10% of the standard.

10. Temperature
The probe calibration will be performed per the SOP annually.

11. Relative Humidity (RH)
The probe calibration will be performed per the SOP annually.

12. Precision Checks

a. Gaseous Analyzers - A reference precision value is established with an "on-site" precision source. This is done by assigning a value to the precision source concentration equal to the calibrated analyzer response between 0.005 and 0.500 parts per million (ppm) for SO_2, NO_2, and O3, and between the prescribed range of 0.5 and 5 ppm for CO monitors. If the mean or median concentrations are below the method detection limit of the instrument, the lowest concentration in the prescribed range that can be practically achieved will be selected. Each analyzer is checked for precision at least once every week before zero and span checks are performed and analyzer adjustments are made. The precision concentration is manually introduced at the beginning of the sample train, including inlet line and inlet filter, until a stable digital display trace is observed (about 15 minutes). The indicated value is compared to the reference value by determining the percent difference, diff = $(Value_{meas} - Value_{audit})/Value_{audit} \times 100$. These one-point QC checks should be ≤ 7% for O_3; ≤ 10% for CO and SO_2; ≤ 15% for NO_2, otherwise the analyzers may require recalibration. Calculated annually, these percent differences provide estimates of annual precision and bias. EPA's precision data quality objective

(DQO) is a coefficient of variation ≤ 7% at a 90% confidence level (90% CL CV ≤ 7%) for ozone and 90% CL CV ≤ 10% for CO and SO_2; 90% CL CV ≤ 15% for NO_2. The bias DQO is a 95% confidence level absolute bias ≤ ± 7% for ozone (95% CL ≤ ± 7%) and for 95% CL ≤ ± 10% for CO and SO_2; 95% CL ≤ ± 15% for NO_2.

b. PM_{10} – Precision data are collected with collocated samplers at a given site in DAQs network. Please refer to the Network Plan for the sites with collocated samplers and frequencies. Calibration, sampling, analysis, verification/validation procedures, and method designation must be the same for collocated monitors. The annual precision DQO for collocated PM_{10} sample concentrations is a CV ≤ 10% for all samples with a minimum concentration of 3µg/m^3.

c. $PM_{2.5}$ – Precision data are collected with collocated filter-based samplers. Please refer to the Network Plan for the sites with collocated samplers and frequencies.
The annual precision DQO for collocated $PM_{2.5}$ filter-based and continuous sample concentrations is a CV ≤ 10% for all samples with a minimum concentration of 3µg/m^3.

13. Working Standard Certification

a. Compressed Gas Cylinders – Calibration and audit gas cylinders are certified against NIST reference material once every year. Sample tanks are certified by triplicate analysis using the method *EPA Traceability Protocol for Assay and Certification of Gaseous Calibration Standards*, May 2012. An unknown sample tank concentration generated by the dilution method is introduced into an uncalibrated analyzer at approximately 80% of the normal operating range until a stable indicated value is achieved. An apparent sample tank concentration is calculated from the measured sample and dilution flows and indicated concentration level. Then, a known NIST SRM concentration is introduced into the analyzer also at approximately 80% of full scale until a stable indicated value is achieved. A true sample tank concentration is derived by multiplying the ratio of the known NIST concentration to the indicated NIST concentration by the apparent sample tank concentration. This procedure is repeated in triplicate analysis. The true sample tank concentration is the average concentration from the three runs.

b. PM Flow Transfer Standards (FTS) – Volumetric flow transfer standards (FTS) used to audit $PM_{2.5}$, PM_{10}, and $PM_{10\text{-}2.5}$, analyzers are certified once each year against a NIST traceable flow standard by the manufacturer of the flow device. The FTS must be accurate within ± 2% of the NIST traceable standard.

c. UV Photometer Transfer Standard – The primary photometer is delivered annually to the EPA Regional lab for certification. The remaining photometers are calibrated quarterly against the primary photometer.

d. Barometers, Thermometers, and Laboratory Balance Weights: Barometers, thermometers, and laboratory balance weights are certified annually against NIST traceable standards. Certification of the filter robot thermometer is performed by the robot manufacturer annually. The laboratory balance weights are sent to a certified weights and measures laboratory for certification. Each certification is accompanied by a signed certificate listing the measurement tolerances.

14. Ambient Data Traceability to NIST and Other Acceptable Standards

All valid ambient data must be traceable to NIST standard reference material as specified in 40 CFR Part 58. Parameters not covered under 40 CFR Part 58 must be traceable to generally accepted ambient air monitoring working standards. This will assure that ambient

pollutant concentration and meteorological parameters are measured with a reasonable degree of accuracy and precision. DAQ has established data traceability using the following criteria:

a. Calibration and audit standards are referenced to NIST SRMs except for ozone transfer standards which can only be referenced by comparison to a UV photometer designated as a primary standard. For meteorological parameters, only those methods and equipment are used as standards, which have a good record for accuracy and reliability.
b. Calibrations and performance audits are performed with certified calibration systems and calibration standards.
c. Span and precision checks are performed with NIST traceable working standards, except for ozone, for which the span source is traceable to a designated primary UV photometer.
d. Zero air for zero checks is traceable to generally accepted scrubbing materials removing moisture and interferents from the ambient air.
e. Laboratory equipment and materials are certified to NIST traceable standards, if available.
f. Inter-laboratory tests are performed with NIST traceable standards and certified equipment.
g. Inter-laboratory calibration system comparisons are performed with NIST traceable standards and certified equipment.

15. Calibrator Side-by-Side Comparison

Every three months all field calibration systems used to calibrate and audit gaseous parameters are compared to a stationary calibrator designated as number 1 that has conferred authority to generate true pollutant concentrations. Side-by-side comparisons are performed by the calibrators who operate and maintain the calibration systems permanently assigned to them. Calibration systems are set up in the Technical Center maintenance is performed prior to the comparisons. Concentrations at approximately 80% full scale are generated for a parameter with each calibration system.

Concentrations are introduced in random sequence into the analyzer until a stable reading is obtained (approximately 20 minutes). The concentrations generated by the field audit calibration systems and the stationary calibrator are compared and a percent deviation is calculated for each field audit calibration system. The percent differences must be within ± 4.0% of the stationary calibrator designated number 1.

Working Standard Certification

Gaseous	**Frequency**	**Acceptance Limits**
CO Tanks	12 months	± 2% of NIST Standard
NO Tanks	12 months	± 2% of traceable NIST Standard
SO_2 Tanks	12 months	± 2% of traceable NIST Standard
Primary O_3 Standard (certification/recertification to SRP)	Once per year	Single point difference ≤ ± 3%
Calibrator side by side comparison	Every 3 Months	
Particulate	**Frequency**	**Acceptance Limits**

PM_{10} FTS	Once per year	± 2% of traceable NIST Standard
$PM_{2.5}$ FTS	Once per year	± 2% of traceable NIST standard
Thermometers	Once per year	± .0.1°C resolution, ± 5 mm Hg accuracy, returned to manufacturer for certification
Barometers	Once per year	± 1 mm Hg resolution, ± 5 mm Hg accuracy, returned to manufacturer for certification
Primary mass standards (compared to NIST-traceable standards)	Once per year	Equal to or less than 25µg
Working mass standards (compared to primary mass standards)	Every 3 months	25 µg

16. Inter-Laboratory Tests
DAQ participates in inter-laboratory studies for PM_{10}, $PM_{2.5}$, NO_2, and SO_2. These studies are initiated by EPA as part of a nationwide assessment for data accuracy. Participation in other parameter tests is anticipated as soon as they are made available by EPA. Samples of unknown pollutant concentrations are sent to the Technical Center for analysis at time intervals determined by mutual agreement.

Samples of unknown concentration (blind audits) are analyzed by the same method and analyzers deployed for routine monitoring. Determined concentrations are submitted to EPA, who calculates a percent error between the submitted concentrations and the true sample concentrations. EPA returns the results together with a histogram showing how DAQ performed on the national test with respect to other agencies.

17. Inter-Laboratory Primary Ozone Standard Certification
Annually, DAQ's primary UV photometer is certified against EPA Region 8's NIST traceable UV photometer. The unit is shipped to the EPA laboratory in Denver. The regional photometer was previously certified against a Standard Reference Photometer from the EPA national laboratory.

For the purpose of comparison, values determined with regional photometer are assumed to be true. Any deviation from these values is expressed as a percent deviation from the true values. Single point difference must be ≤ ± 3%; otherwise, DAQ's photometer must be recertified.

18. Data Accuracy and Precision Calculations
As required by 40 CFR Part 58, Appendix A, DAQ must submit precision and accuracy data to AQS within 90 days of the end of the quarter. The quarterly submittals are summarized in AQS Report AMP 247 (Accuracy Report). For each CAP, percent difference is calculated for each single point check between the measured and the audit concentration or between a measured and audit set point flow rates. It is the responsibility of the QA Officer to calculate these statistics, to compare the calculated results against the Measured Quality Objectives (MQO) listed in Table 1 of Appendix A, 40 CFR 58, and to inform the Monitoring Section manager if the MQOs are being achieved.

Working with staff, the manager should identify the major sources of uncertainty in the audits and find ways to reduce the uncertainty to acceptable levels.

19. Performance and Traceability Tests for Digital Volt Meters (DVMs)
DAQ uses modern DVMs that measure voltage, current, and resistance. Each DVM's performance is tested annually by designated monitoring staff. One DVM with the highest rated accuracy (preferably 4 1/2 digits; Fluke 8020A or equivalent) is sent to an instrument laboratory for calibration against a NIST traceable standard. This most recently calibrated DVM becomes the reference against which the other DVMs are checked. Tests are conducted for several specified ranges of AC and DC voltages, currents, and resistance. If the indicated voltages, currents, or resistances are outside of the acceptable accuracy range, the DVM must be adjusted to meet the acceptable range. If this is not possible, it must be repaired and recalibrated by an authorized service laboratory. DVM tolerance must be ≤ ±0.3% ± one digit. If the DVM is outside of tolerance limits, it must be adjusted or sent to an instrument laboratory for repair and calibration.

DVM Certification Target Ranges

DVM Function	1st Target Range	2nd Target Range	3rd Target Range	4th Target Range
D.C. voltage	100 mV	5.0 V	30 V	
A.C. voltage	5.0 V	100 V	120 V	
D.C. current	0.5mA	100mA		
A.C. current	0.5mA	100mA		
Resistance	10 Ω	1.0K Ω	100 KΩ	1 MΩ

20. Reporting Assessment Information
Assessment information for PM_{10}, and $PM_{2.5}$ are reported to the data processing team who review and summarize the data into precision and accuracy reports that are uploaded in AQS. When errors are observed, they turn to the field operators who generated the data and request that the audit/calibration be repeated. The team also reviews the raw data looking for outliers and data flags as they input it into AQS. When particulate data entry into AQS requires corrective action, the field operator who was assigned the station, is requested to provide documentation.

Assessment checks of gaseous parameters are initially reported to the validation team. Similar to the submitted particulate data, the validation team reviews and summarizes the data into precision and accuracy reports that are uploaded in AQS. Concurrently, they validate and code the data, checking on outliers and unexplained data gaps. When corrective action is required, they question the field operator responsible for the data, requesting documentation when required. They enter the gaseous and continuous particulate data into AQS.

The QA Officer may receive and review assessment reports. If precision and bias estimates show large percent differences between measured and audit values over a week or longer, the field operator must investigate underlying causes for the trend (i.e., a UV ozone-source lamp requires replacement, poor filter handling techniques, etc.). If the signed bias is always positive or negative, systematic error in the measurement is suggested and must be investigated. If the statistical analyses appear to show that certain monitors are trending

towards out-of-control limits, the field operator needs to investigate the situation and suggest possible fixes to the problem. If the observed problems appear widespread, the subject and possible solutions should be discussed in regularly scheduled staff meetings.

Reports to Management

Reports to management are useful in summarizing data quality of air pollutant measurements. Quality assurance reports are periodically provided to management regarding spot checks of systems and QC. Recommendations for further QC follow through and performance improvements are made. Information for the quality assurance reports will be taken from AQS, AirVision control charts, MTL QA reports, and results from systems and performance audits.

Corrective action reports have generally taken the form of informal verbal reports discussed in section staff meetings rather than a written corrective action request. A corrective action report may be initiated by anyone in the monitoring section who observes a problem that requires either immediate or long-term action to correct a safety defect, an operational issue, or failure to comply with the written SOP. We are currently evaluating the feasibility of creating an online operations, maintenance, and corrective actions log.

Occasionally, DAQ management directs that special studies be conducted in various parts of the State in response to public complaints or federal concerns. These studies are short-term efforts with limited data quality objectives; i.e., perform a saturation study in an area to provide justification for siting a permanent ozone-monitoring station, or monitoring for $PM_{2.5}$ during wintertime temperature inversions to show compliance, or lack thereof, with the federal standard.

A study plan is drafted by monitoring section personnel assigned to the special study or the Principal Investigator (PI). The plan includes an approval sheet that must be signed by section, branch, and division management, a description of the project, project organization and responsibilities, data quality objectives, sampling procedures, testing, installation, and calibration of deployed air quality and meteorological instruments, data reduction and analysis including QA/QC reports, and the final report.

The final special study report adheres to the study plan's outline summarizing the study's findings in the executive summary (report abstract), a description of methods and procedures used (experimental design), results and discussion, conclusions, and recommendations for future work. Appendices to the report should include a detailed description of sites with accompanying sketches and/or pictures, standard operating procedures, and if appropriate, the raw data collected. The report is distributed to section managers and the division director. It is left to the discretion of the division director whether the report is made available to the public.

Data Review, Verification, and Validation

Data review, verification, and validation are techniques used to accept, reject, or qualify data in an objective, consistent manner. The EPA defines data verification as the process of evaluating the completeness, correctness, and conformance/compliance of a specific data set against the method, procedural, or contractual requirements. Data verification is the responsibility of the field operators who visit the stations and collect data according to DAQ's SOP, who ensure that the sample ran the correct date and time frame, who check the monitor's operating parameters (flow, pressure, temperature, etc.) to document that it ran properly. Data validation is defined as an analyte and sample-specific process to determine the analytical quality of a specific data set.

Data is considered valid or invalid based on established criteria as explained below. Data flagging is only done in cases where anomalous event(s) occur.

Verification and Validation Methods

Collected data is stored in the station's data logger. The logger invalidates data where power outages, calibrations, audits, maintenance, etc., occur. This automated program codes for some invalid data. Independent data validators review all data for invalid data, data gaps, etc., and attach the requisite data codes before entering the data into AQS.

Data is accepted as valid for entry into the AQS database if the following conditions are met:

1. All data should have been validated according to established criteria contained in the Quality Assurance Project Plan and procedures manual.
2. All pertinent documentation must be available and verified for data traceability.
3. Data accuracy for gaseous analyzers should fall within established limits based on weekly spans and daily zeros; for PM_{10} and $PM_{2.5}$ on monthly verification audits for flow rate, ambient temperature, and ambient pressure; and annual audits for meteorological parameters.
4. All questionable data must be evaluated and may be flagged from the working data files. Non-routine data issues identified by the independent validators must be investigated and documented in a QA memo. The QA memo shall describe the event, resolution and be approved by the QA Officer and Monitoring Section Manager.
5. AQS data should be compared against the internal database as a final accuracy and completeness check.

The Air Monitoring Section Manager is the person authorized to make the final decision on the validity of the data and to make changes as needed based on recommendations from the QA officer, the independent data validator, and the Technical Analysis Section Manager.

Data Validation - Gaseous Data
Conditions and Limits for Validating Data

Condition	Limits	References/Comments
Zero Drift	CO 0.41 ppm	EPA Handbook, Vol II, App D
	O_3, NO_2, $SO_2 \leq \pm 0.0031$ppm	
Zero Checks	No more than 2 weeks between checks	Acceptable time between instrument performance checks
Precision/Span Drift	$O_3 \leq 7\%$, CO, SO_2 each $\leq \pm 10\%$, NO_2, NO_y, $\leq \pm 15\%$	EPA Handbook, Vol II, App D
Span Checks	≤ 2 weeks between checks	Acceptable time between instrument performance checks
Data Capture	≥ 45 minutes/hour	Quality Assurance Handbook for Air Pollution Measurement Systems Vol. I, Principles
Station Temperature	20-30°C. (Hourly average)	Generally, the 20-30°C range will apply but the most restrictive operable range of instruments in the shelter may be used as guidance
No excessive Instrument Spiking	Assessed on a case-by-case basis	Instrument performance okay
Instrument is in ambient mode when sampling	N/A	
Inlet line connected during sampling	N/A	
Daily operational status log entry at least every other day	N/A	
Power outages	1-hour warm-up time for each hour of power outage	To make sure instrument reaches proper operating temperature
CO, NO_2, SO_2: Gas dilution systems	Accuracy ± 2%, checked once every three months	EPA Handbook, Vol II, App D

Data Validation - PM_{10}
Condition and Limits for Validating Data

Condition	Limits	Comments
Average flow rate ─ for every 24 hr period	Average within ± 5% of 16.7 ℓpm (sampler set point)	QA Handbook, Vol II, App D
Flow rate variability rate ─ every 24 hrs of operating period	CV ≤ 2%	QA Handbook, Vol II, App D
Sample Recovery	≤ 7 days 9 hours from sample end date	QA Handbook, Vol II, App D
Post-Sampling Weighing	≤10 days from sample end date if shipped at ambient temp, or ≤30 days if shipped below average ambient (or 4°C or below for average sampling temps <4°C) from sample end date	QA Handbook, Vol II, App D
Filter damage	No evidence of filter damage	QA Handbook, Vol II, App D
Sampling Period	Sampler operated at least 23 but no more than 25 hours	QA Handbook, Vol II, App D
Filter Equilibration Temp. range/variability	24 hours minimum 24-hr mean: 20-23°C, ± 2°C SD	QA Handbook, Vol II, App D
Filter Equilibration Humidity range/variability	24 hr mean 30% - 40% RH, ± 5% SD over 24 hours	QA Handbook, Vol II, App D
Timer accuracy	± 30 minutes of midnight	Reference method specification
Information on the data card	Information is complete	Example: off flow, gross weight, etc.
Availability of data card	Data card is not lost or misplaced	Each filter must have a completed, corresponding data card
Lot filter blanks	< 15 µg change between weighing	QA Handbook, Vol II, App D
Exposed filter integrity	No visual defects	QA Handbook, Vol II, App D
Filter holding times: pre-sampling	< 30 days before sampling	QA Handbook, Vol II, App D
Field filter blank	Optimally, ± 30µg change between weighing. Historical data indicates greater variability is common.	Optimally, 10% or 1 per weighing session. Investigate major deviations from norm.
Lab filter blank	± 15µg change between weighing	10% or 1 per weighing session
Balance check	≤ 3µg	Every 10th sample weighed
Duplicate filter weighing	± 15µg change between weighing	10% or 1 per weighing session

Data Validation - $PM_{2.5}$
Conditions and Limits for Validating Data

Condition	Limit	Comments
Average flow rate ─ every 24 hrs of operation	Average within ± 5% of 16.7 ℓpm (sampler set point)	QA Handbook, Vol II, App D
Flow rate variability ─ every 24 hrs of operation	CV ≤ 2%	QA Handbook, Vol II, App D
Sample recovery	≤ 7 days 9 hours from sample end date	QA Handbook, Vol II, App D
Post-Sampling Weighing	≤10 days from sample end date if shipped at ambient temp, or ≤30 days if shipped below average ambient (or 4°C or below for average sampling temps <4°C) from sample end date	QA Handbook, Vol II, App D
Sampling Period	Sampler operated at least 23 but no more than 25 hours	QA Handbook, Vol II, App D
Filter Damage	No evidence of filter damage	QA Handbook, Vol II, App D
Filter Equilibration Temp. range/variability	24-hr mean: 20-23°C, ± 2°C SD	QA Handbook, Vol II, App D
Filter Equilibration Humidity range/variability	24 hr mean 30% - 40% RH, ± 5% SD over 24 hours	QA Handbook, Vol II, App D
Pre-/post-sampling RH	Difference in 24-hr means ≤ ± 5% RH	QA Handbook, Vol II, App D
Information on data cards	Information is complete	Example: off flow, gross weight, etc.
Data card availability	Data cards are not lost or misplaced	Each filter must have a completed, corresponding data card
Flow Blockage	"Off flow" is within ± 10% of the set point flow	
Lot filter blanks	< 15 µg change between weighing	QA Handbook, Vol II, App D
Exposed filter integrity	No visual defects	QA Handbook, Vol II, App D
Filter holding times: pre-sampling	< 30 days before sampling	QA Handbook, Vol II, App D
Field filter blank	Optimally, ± 30µg change between weighing. Historical data indicates greater variability is common.	Optimally, 10% or 1 per weighing session. Investigate major deviations from norm.
Lab filter blank	± 15µg change between weighing	10% or 1 per weighing session
Balance check	≤ 3µg	Every 10th sample weighed
Duplicate filter weighing	± 15µg change between weighing	10% or 1 per weighing session

Data Validation - Meteorological
Condition and Limits for Validating Data

Condition	Limits	Comments
Met Sensor Alignment Audit	All alignments are ≤ ± 2 degrees	If landmarks or direction markers are used
Ambient Temperature Audit	± 0.5°C accuracy, 0.1°C resolution	NA
Dew Point Temperature	≤ ± 1.5°C (2.7°F) of the dew point temperature, 0.1°C resolution	QA Handbook for Air Pollution Measurement Systems, Vol IV, Appendix C (May 2013)
RH	± 7% (accuracy), 0.5% resolution	QA Handbook for Air Pollution Measurement Systems, Vol IV, Appendix C (May 2013)
Barometric Pressure	± 3 mb Hg accuracy, 0.5 mb Hg resolution	QA Handbook for Air Pollution Measurement Systems, Vol IV, Appendix C (May 2013)
Audit Frequency	Audit must have been conducted in previous 12 months	QA Handbook for Air Pollution Measurement Systems, Vol IV, Appendix C (May 2013)
Up Time	Each hour must contain at least 45 minutes of data	Quality Assurance Handbook for Air Pollution Measurement Systems, Vol I, Principles
Data System Sampling Frequency	System must sample wind speed, wind direction, and temperature at least once per second	All modern data loggers sample once each second

Reconciliation with Data Quality Objectives and User Requirements

Data objectives are reconciled by:

a. Evaluating ambient pollutant concentrations with respect to National Ambient Air Quality Standards (NAAQS) at a representative measurement scale, e.g., micro-scale, middle-scale, neighborhood scale, urban scale, etc.
b. Demonstrating that an area is attainment or non-attainment with NAAQS.
c. Having monitoring sites located in representative high concentration locations in non-attainment areas as defined by modeling, saturation studies, and traffic counts.
d. Accurately tracking ambient pollution concentrations in support of a State Implementation Plan.
e. Correlating ambient pollutant concentrations with reported emission reductions or increases.
f. Designating an area with its appropriate attainment classification based on its determined design value.
g. Collecting enough data to assess the air quality and air quality trends within the boundaries of the State.
h. Providing air pollution data to the general public in a timely and accurate manner to adequately characterize an area's pollution levels for air pollution index reporting.

2. Reconciliation with user requirements**:**
   Any ambient monitoring data that fails to meet the requirements found in 40 CFR, Parts 50, 53, and 58 and their applicable appendices, and the conditions of this Quality Assurance Project Plan may be used for regulatory purposes by applying a weight-of-evidence review by the supervisor and QA Officer according to the process describe in D2 above.

The procedures used to assess precision, bias, and accuracy for the ambient air quality data are contained in 40 CFR Part 58, Appendix A. These statistical estimates provide measurements of uncertainty in the validated data. DAQ has calculated precision and absolute bias estimates for the gaseous and particulate parameters annually and tri-annually beginning with data from 2007 forward. Failing to meet these EPA-published data quality indicators casts suspicion on the data and the reason for not meeting the criteria must be investigated and minimized to reduce the likelihood that additional samples may be invalidated. The procedure used to assess data completeness is contained in the *Quality Assurance Handbook for Air Pollution Measurement Systems, Volume II, Appendix D (revised May 2013).*

Data Management Security

DTS, our IT department controls access to the AirVision Server. The AirVision client access is maintained by the air quality network administrator. When a new user is added to the AirVision database, they are put into one of four groups: Report (read-only access), User (air quality staff), QC (air quality staff that do QC), and Admin (all access). When a user retires or leaves, their account password is changed or deleted. If a user stops using AirVision and does not get their software updated when the server software is upgraded, they lose access.

Photochemical Assessment Monitoring Stations (PAMS)

Background

The Photochemical Assessment and Monitoring program is a subset of the SLAMS network that is being implemented for enhanced monitoring of ozone precursor chemicals at NCore sites located in an area with a population over 1,000,000 and in areas of moderate and above nonattainment status. Required PAMS measurements at the NCore site include volatile organic compounds (VOCs) measured by automated gas chromatograph (AutoGC), carbonyl compounds, O3, NO, NO_2, NO_y, and meteorological parameters including ambient temperature, wind direction, wind speed, barometric pressure, relative humidity, precipitation, mixing height, solar radiation, and ultraviolet radiation. Other sites required as part of an enhanced monitoring plan (EMP) for other areas that are non-attainment for ozone may have some or all of these criteria or different measurement criteria. DAQ is implementing PAMS for the full calendar year to account for both summer and winter ozone seasons.

This document provides an overview of the work to be conducted, defines the method and establishes the QA/QC activities that will be applied in order to attain the developed objectives.

Project/Task Description

Required Measurements: In order to fulfill the requirements listed in 40 CFR Part 58 Appendix D Section 5, UDAQ will measure:

- Hourly averaged concentrations of priority speciated VOCs and TNMOC
- Three sequential 8-hour average concentrations of carbonyls determined on a 1-in-3-day schedule;
- Hourly averaged concentrations of O_3 (included in the network QAPP document);
- Hourly averaged concentrations of nitrogen oxide (NO), “true” nitrogen dioxide (NO_2), and total reactive nitrogen (NO_y) (NO and NO_y are covered in the network QAPP);
- Hourly averaged ambient temperature;
- Hourly vector-averaged wind direction;
- Hourly vector-averaged wind speed;
- Hourly average atmospheric pressure;
- Hourly averaged relative humidity (RH);
- Hourly precipitation amount;
- Hourly averaged mixing layer height (MLH);
- Hourly averaged intensity of solar radiation.

Data Quality Objectives and Criteria

Completeness

Method or Parameter	DQI				
Parameter	**Representativeness (Sampling Frequency)**	**Bias (%)**	**Precision (%)**	**Sensitivity (Detection Limit)**	**Completeness (%)**
Auto-GC speciated VOCs	Valid ambient recovery adjusted for QC Continuous, hourly average	≤ ±30 for benzene or toluene (depending which is the calibration for the BP column) propane ≤ ±30 for the other analytes	≤ 25	≤ 0.5 ppbC or MDL	≥ 75
True NO_2		< ±15.1%	< 15.1%	≤ 0.001 ppm	≥ 75
TO-11A (carbonyls)	Three sequential 8-hour samples every 3rd day	NA	NA	NA	≥ 75
Meteorology	**Representativeness (Sampling Frequency)**	**Bias**	**Precision**	**Sensitivity (Resolution)**	**Completeness (%)**
Ambient Temperature	Continuous, hourly average	≤ ± 0.5 ºC	not specified	≤ 0.1 ºC	≥ 75
Relative Humidity		≤ ± 5% RH ≤ ± 1.5 ºC of the dew point temperature		≤ 0.5% RH	
Barometric Pressure		≤ ± 3 hPa		≤ 0.1 hPa	
Wind Speed		± 5%,		≤ 0.1 m/s	
Wind Direction	Continuous, hourly average	≤ ± 5 degrees	not specified	≤ 1 degree	≥ 75
Solar Radiation		≤ ± 5% during sunny conditions		≤ 1 Watt/m²	
UV Radiation		≤ ± 5% during sunny conditions		≤ 0.01 Watt/m²	
Precipitation		≤ ± 10%		≤ 0.25 mm/hr	
Mixing Layer Height		≤ ± 5 m		NA	

Carbonyl make-up testing: the make-up testing requirement will only be fulfilled if the operator is aware of a missing sample, in that case, the EPA-contract laboratory will be contacted to request a makeup cartridge for replacing the missing sample.

Precision

Precision for ceilometer: NA

Precision for Continuous Measurements of true NO_2: Precision for true NO_2 is assessed by calculating the upper 90% confidence limit (CL) for the coefficient of variation (CV) across all single-point QC checks measured annually (or during PAMS monitoring if NO_2 monitoring is limited to PAMS season). For the functional form of the calculation, see 40 CFR 58 Appendix A Section 4.1.2, Equation 2.

Precision for Speciated VOCs: Precision is assessed by analysis of replicate calibration verification standards to evaluate the reproducibility of the analysis. A weekly sequential

(back-to-back) analysis of the continuing calibration verification (CCV) is performed for the speciated VOCs precision check.

Within-run precision is evaluated by comparing replicate measurements in a pair-wise fashion to determine the relative percent difference (RPD) between the two measurements which must be ≤ 25% for each assessed target compound in the calibration verification standard (CVS). Ambient sample data for target analytes failing precision criteria will be appropriately qualified or invalidated, as appropriate, when reported to AQS.

Precision for Carbonyls: Precision will not be calculated because duplicate samples are not collected.

Bias

Bias is the systematic or persistent distortion of a measurement process that causes error in one direction. It is estimated from the positive and negative deviation from the true value as a percentage of the true value.

For speciated VOCs, the bias is evaluated by ensuring that the percent recoveries of the daily calibration verification standard (CVS) and weekly laboratory calibration standard (LCS) as well as the daily method blank concentrations remain within the specified tolerances in the SOP (AMS-OPS-001 "*Automated Gas Chromatograph Data Collection, Management, Validation, and Reporting*").

For carbonyls, bias may occur from a variety of field activities and is assessed and minimized by a combination of pre-deployment bias checks; such as, flow verification for sample collection instruments. Sampling unit flow rates will be verified before field deployment and monthly thereafter against a NIST traceable certified reference flow transfer standard.

Bias of true NO_2 measurements is assessed by conducting an annual performance evaluation (PE) described in 40 CFR Part 58 Appendix A, Section 3.1.2. Additional PE assessments may be conducted minimally every 5 years by EPA as part of the National Performance Audit Program (NPAP) as described in 40 CFR Part 58 Appendix A, Section 2.4.

General Training Requirements

Site operators require special training to calibrate, operate, maintain, and troubleshoot instruments and support equipment needed to make PAMS measurements. Each site operator should possess the appropriate skills and education to perform their assigned tasks. Instrument vendors may train site operators on how to set up, calibrate, and operate auto-GCs, carbonyls samplers, and ceilometers during installation. Site operators will receive training materials and will attend training sessions and online training webinars provided by instrument vendors, EPA, and/or experts as they are available and resources permit.

Documentation and Records

PAMS VOC and carbonyl data will be submitted to AQS as expeditiously as possible. The raw data and validation will be archived for a minimum of 5 years.

Schedule of Deliverables to EPA

PAMS data should be delivered within 180 days of the end of the calendar quarter in which the measurement data were collected. ALS prepares, distributes sampling media and analyzes dinitrophenylhydrazine (DNPH) coated silica cartridges for carbonyls. In addition, it validates and reports the data to the AQS.

Activity	Start Date	Completion Due Date	Details
Data Reporting to AQS	Not before data validation is completed	Within 180 days of the end of the calendar quarter in which the measurement data were collected and the data has been validated	Input of validated measurement data to AQS. Upload of all PAMS parameter data to AQS and verification that data were uploaded correctly

Data Generation

Sampling Process Design

The PAMS program is designed with the objective to produce an air quality database to be used in the ozone prediction model evaluation and refinement. In addition, the program will assist to identify and quantify the ozone precursors, establish the temporal patterns and associated meteorological conditions to assist and refine the control strategies. DAQ will measure the following parameters at the PAMS required site:

- Carbonyls
- Meteorological parameters: ambient temperature, wind direction, wind speed, atmospheric pressure, relative humidity, precipitation, mixing layer height and solar radiation
- Speciated VOCs
- True NO_2
- NO/NO_y
- Ozone

Sampling Collection Schedule

Parameter	Sampling Duration and Frequency	Value Reported
True NO_2	Continuously, daily	Hourly average
NO/NO_y	Continuously, daily	Hourly average
Carbonyls	3 sequential 8-hour samples on a 1-in-3-day schedule (June 1-August 31), 1 in 24-hr every 3-days (remaining part of the year)	8-hour average
Speciated Volatile Organic Compounds (VOC), Continuous	Hourly 40-minute sample, daily	Hourly average

Sampling Methods Requirements

Standard operating procedures for ceilometer, true NO_2, NO_y, VOC auto-GC, and carbonyls are contained in the Utah Ambient Air Monitoring Procedure Manual.

Parameter	Manual or Automated Reference/Equivalent Method
Nitrogen Dioxide (NO_2)	EQNA-0320-256 Teledyne Advanced Pollution Instrumentation, Inc. Model N500
Reactive Oxides of Nitrogen (NO_y)	RFNA-1194-099 Teledyne Advanced Pollution Instrumentation, Inc. Model T200U/NO_y Analyzer
Carbonyls	Silica DNPH Cartridge with KI ozone scrubber collected using ATEC 8000 sampler and analyzed with HPLC UV Absorption
Speciated Volatile Organic Compounds (VOC), Continuous	Markes CIA Advantage-xr sampling preconcentrator, Kori-xr electronic dryer and UNITY-xr thermal desorber with Agilent 7890b GC – carbon response

Sample Handling and Custody Requirements

Carbonyls by TO-11A

Sample Handling:

New cartridges will be stored at ≤ 4°C upon receipt and be maintained sealed in their original packing and protected from light until installed for sample collection. The following procedure will further limit potential contamination:

- Cartridges will only be handled with gloved hands (chemicals from hand sanitizers, lotions, etc., can contaminate sample cartridges).
- Markers containing volatile solvents (e.g., permanent markers such as Sharpie®, which emit solvents such as acetone) will not be used for marking on foil storage pouches. Ball-point pens or printed stick-on labels are preferred.
- Cartridges will be labeled in such a manner to uniquely identify the cartridge, i.e., to permit the identification of the sampling date, time, and whether it is a primary sample or field QC sample (field blank).
- Cartridge storage areas will be free of carbonyls. Climate-controlled storage units will not be used for storage of solvents or carbonyl-containing solutions or standards.
- Upon retrieval, each cartridge will be capped, sealed in its individual foil pouch, and placed immediately in cooler with ice packs. Collection details will be documented appropriately on the sample collection form (whether hard copy) and sample storage information will be recorded on the chain of custody form. Samples will be hand-carried or shipped by overnight courier/shipper to ensure that samples arrive at the laboratory under refrigeration.

Holding times:
Samples are retrieved, capped, protected from light, and stored at ≤ 4°C within 72 hours of end of 3rd sequential sample (whose sampling is nominally completed at 4:00 a.m.).

Pollutant	Criterion	Acceptable Range
Carbonyls	Duration from end of sample collection to sample extraction	≤14 days from end of sample collection

Sample Control
The following information must minimally be recorded on the control form:

- Origin of cartridges (e.g., analysis laboratory or field office)
    - Receipt of field-collected cartridges by analysis laboratory
- Unique identifier(s) for each sample, sample collection date(s), and site(s) location information
- Storage of cartridges at each point during transfer between individuals, including during shipment
    - Storage of field-collected cartridges at the monitoring site, if applicable (e.g., stored at ≤ 4°C in an onsite refrigerator, etc.)
    - Shipping conditions (e.g., on ice packs) and associated information for tracking or evaluating the shipping conditions - such as thermometers placed in a shipping cooler
    - Upon receipt at the laboratory – document thermometer used for measuring temperature as received and location for storage within laboratory (e.g., uniquely identified refrigerator).

Laboratory sample custodians, or designated individuals responsible for assuring sample custody, will ensure that sample custody documentation is complete. Site operators will be contacted, as appropriate, to complete missing information. COC documentation will be maintained in accordance with Section B.3 of Utah Network QAPP.

Analytical Methods

All of the analyzers and sampling methods for the criteria pollutants used by the DAQ are reference or equivalent methods.

Measurement Parameter	Manual or Automated Reference/Equivalent Method
Speciated VOCs	Auto-GC with thermal desorption and dual FID
Carbonyls	Carbonyls-specific sampler capable of three sequential 8-hour samples – Collection and analysis per EPA Compendium Method TO-11A. The ASL will require a high-performance liquid chromatograph (HPLC) or ultra-high-performance liquid chromatograph (UHPLC)
Ozone	Federal Reference Method/Federal Equivalent Method (FRM/FEM) ozone analyzer
True NO_2	FEM CAPS NOx analyzer is a Cavity Attenuated Phase Shift (CAPS) Nitrogen Oxides Analyzer
Ambient Temperature	Thermistor
Wind Direction	Sonic anemometer
Wind Speed	
Atmospheric Pressure	Pressure transducer
Relative Humidity	Relative humidity transducer
Precipitation	Tipping bucket
Mixing Layer Height	Ceilometer
Solar Radiation	Pyranometer
Ultraviolet Radiation	Pranometer

ASL-analytical support laboratory

Carbonyls by TO-11A

Samples collected for carbonyls analysis will be extracted and analyzed per EPA Compendium Method TO-11A and will meet the performance specifications established in the Compendium Method. This method is described in the national SOPs for the collection of analysis of carbonyls for the PAMS Required Site Network, which describe the procedures for solvent extraction of derivatized carbonyl-hydrazones collected on the DNPH cartridge samples, analysis of these extracts by HPLC or UHPLC with UV detection, and the necessary QC procedures. Ambient air and QC samples must be extracted for analysis within 14 days of collection. Extracts are then analyzed by HPLC or UHPLC with UV at 360 nanometers (nm) within 30 days of extraction and the carbonyl concentrations in the ambient air sample calculated from the measured concentrations in the sample extracts and the volume of air sampled onto the cartridge. Alternative detectors (such as time-of-flight [TOF]) and alternative wavelengths (e.g., 365 nm) may be employed, if method performance criteria (listed in Table B5-5) are met.

Speciated VOCs by Auto-GC

Auto-GC systems will be used for the analysis of VOCs. The auto-GC systems collect and preconcentrate VOCs from the ambient atmosphere and subsequently separate the VOCs for detection by FID. Procedures for the setup, calibration, operation, and shutdown of auto-GCs are described in PAMS Required Site SOPs. Analysis of speciated VOCs by auto-GC is considered a continuous method, therefore there are no discrete samples collected on media which may be retained.

Parameter(s)	Analytical Method Designation
Carbonyls	EPA Compendium Method TO-11A *"Determination of Formaldehyde in Ambient Air Using Adsorbent Cartridge Followed by High-Performance Liquid Chromatography (HPLC)"*
VOCs, TNMHC and TNMTC by auto-GC	PAMS Technical Assistance Document "*Quality Assurance Project Plan for the Photochemical Assessment Monitoring Stations (PAMS) Required Site Network for Speciated Volatile Organic Compounds, Carbonyls, and Meteorological Parameters Including Mixing Layer Height – August 2020"*

True NO2 by FEM

True NO_2 analyzers approved as FEMs will be employed for the analysis of true NO_2. The analyzers continuously sample ambient air routed through an in-line particulate matter (PM) filter, therefore no discrete samples are collected on media which may be retained.

The analyzers are calibrated for NO_2 response (NO_2 is generated by gas phase titration with excess NO) by providing the analyzer with a zero-concentration gas and three of the four concentrtaions in the audit ranges between zero and 80% of the span value. Following, minimally every 14 days thereafter, the analyzer is subject to a zero/span check to verify ongoing instrument calibration.

Quality Control

1. Quality Control Schedules

Carbonyl Mass flow controller calibration verification when the instrument is first received (Verify sampling flow of each channel at the sampling flow setting).

Daily:

a) Auto-GC System blank (Analysis of humidified zero air to ensure the system is sufficiently clean for continued analysis. Prior to ICAL, and every 24 ± 4 hours of operation following the CVS (to ensure absence of carryover before analyzing ambient samples).
b) Auto-GC CVS (Concentration of CVS should be approximately 2-5 ppbC for target analytes).
c) Perform zero checks remotely on continuous gas analyzers. The zero drift for NO_X true $\leq \pm$ 3%.
d) Check stability of station temperature. The shelter temperature range should be maintained between 20-30°C (68-86°F).

Weekly:

a) Auto-GC Second Source calibration verification (LCS).
b) Auto-GC retention time standard (RTS)- Analysis of a 59-component blend of VOCs in the ~2 to 60 ppbC range to verify established retention time windows.
c) Auto-GC precision check - Replicate analysis of the CVS to evaluate the reproducibility of the analysis (replicates are analyzed back to back).

Complete AutoGC site visit and fill out operator check list.

True NO_2 Zero/Span/ Span precision check.

Monthly:

Carbonyl Field blank.

Semi-annually:

Perform multipoint calibrations on criteria continuous gaseous analyzers and Auto-GC.

Annually:

a) Maintenance activity (prior to season start):
   - Auto-GC: Clean or change Kori-xr trap, replacing pre-concentrator trap(s) and servicing zero air generator.
   - Carbonyl: replacing the ozone denuder and particulate filter.
   - True NO_2: replacement of particulate filter.
b) True NO_2 Calibration.
c) Meteorological Sonic instruments will be calibrated using reference instruments per the DAQ Meteorological Instrument Calibration SOP.

2. Zero and Span Checks

Zero checks for True NO_2 analyzer: Verification performed by analyzing a zero and two standard NO_2 concentration levels – span point at approximately 80% of the measurement range and a precision point in the lower 1/3 of the measurement range. Zero drift must be less than ± 0.3 ppb. Span and precision levels must be within ±10% and 15% respectively of their theoretical concentration.

3. Adjustment and Out-of-Control Conditions

Parameter	Condition	Corrective Action
Carbonyls-mass flow controller calibration verification	Carbonyl mass flow calibration verification within ±10% of flow transfer standard	Recalibrate flow controller. If problem persists, investigate for leaks, blockage in flow path, etc. May require replacement of MFC or other instrument parts.
Carbonyls-field blank	Measured mass per cartridge (µg/cartridge): - Acetaldehyde ≤ 0.40 - Formaldehyde ≤ 0.30 - Acetone ≤ 0.75 - Sum of other compounds ≤ 7.0	Investigate sources of contamination in handling and transport. The ALS lab analyzes, validates, and reports the data to the AQS
AutoGC multipoint calibration (ICAL)	The final response factors (RFs) ≤ 10% of the previous RF	Prepare new calibration. It may be necessary to investigate for system contamination or interferences resulting in suppression or enhancement of analytes. System leaks and trap degradation may impede a proper calibration as well as carryover from samples or standards. See SOP (AMS-OPS-001 "*Automated Gas Chromatograph Data Collection, Management, Validation, and Reporting"*) for response action details.
Auto-GC system blank	All compounds < 0.5 ppbC or MDL	Analyze another blank, if possible, to investigate potential carryover from high concentration sample. Investigate system for contamination. See SOP (AMS-OPS-001 *"Automated Gas Chromatograph Data Collection, Management, Validation, and Reporting")* for response action details.
Auto-GC calibration verification standard (CVS)	Propane (which is the calibrant for the PLOT column) and benzene or toluene (depending on the calibrant for the BP column) percent recovery ≥ 70% and ≤ 130%; same for the other compounds ≥ 30% and ≤ 130%	Investigate chromatogram for retention time shifts which may result in peak misidentification. Investigate for instrument contamination resulting in co-eluting peaks. Investigate for system leaks or trap malfunction resulting in low recovery. See SOP (AMS-OPS-001 *"Automated Gas Chromatograph Data Collection, Management, Validation, and Reporting"*) for response action details.
Auto-GC laboratory calibration standard (LCS)	Propane (the calibrant for the PLOT column) and benzene or toluene (depending on the calibrant for the BP column) percent recovery ≥ 70% and ≤ 130%	Investigate for discrepancy between ICAL and LCS. Investigate chromatogram for retention time shifts which may result in peak misidentification. Investigate for instrument contamination resulting in co-eluting peaks. Investigate for system leaks or trap malfunction resulting in

Parameter	Condition	Corrective Action
		low recovery. See SOP (AMS-OPS-001 *"Automated Gas Chromatograph Data Collection, Management, Validation, and Reporting"*) for response action details.
Auto-GC precision check	Target compound are integrated and identified within the established retention time windows	Review previous week's ambient and QC check sample data to evaluate events resulting in retention time shift. May require reassignment or adjustment of retention time windows and reprocessing of data collected since the most recent CVS or RTS.
True NO_2 zero/span precision check	≤ 3% of full-scale (true NO_2) Zero drift must be less than ± 0.3 ppb Span and precision levels must be within ±10% and 15% of the theoretical concentration	Repeat zero and span checks to confirm. Investigate system for contamination, leaks, or other causes of drift. Qualify or invalidate data since the last passing QC check. Perform calibration and multipoint verification.

Calibration Frequency

Parameter	Frequency	Procedure (SOP Manual)
Auto-GC VOC	Annual Multi-Point calibration and if out-of-control or after major repair	SOP (AMS-OPS-001 *"Automated Gas Chromatograph Data Collection, Management, Validation, and Reporting"*)
Carbonyl	Annual before PAMS season	SOP (AMS-OPS-002 "*Carbonyl sampling and instrument Maintenance*")
True NO_2	Annual Multi-Point calibration and if out-of-control or after major repair	SOP (AMS-OPS-002 "*Trace-level, Photolytic, True NO2/NO/NOx Analyzer*")
Ambient Temperature	Annual before PAMS season	QC checks against NIST-traceable standard
Wind Direction	Annually	Conducted by manufacturer
Wind Speed	Annually	Met Instrument Calibration SOP
Atmospheric Pressure	Annual before PAMS season	Conducted by manufacturer
Relative Humidity	Annual before PAMS season	Per SOP
Precipitation	Annual before PAMS season	QC checks against NIST-traceable standard
Mixing Layer Height	Annual before PAMS season	NA
Solar Radiation	Annual before PAMS season	Conducted by manufacturer. QC checks against NIST-traceable standard

The following paragraphs provide a brief description of how the calibrations will be conducted:

1. Automated Gas Chromatograph

   Semi-annually or following calibration verification failure/major repair, the Markes-Agilent AutoGC system is calibrated. A known quantity retention time standard (RTS) is blended with zero air and introduced into the system in 40-minute samples per hour. The multipoint calibration begins with a blank hour to clear the system, then RTS concentrations of approximately 1, 2, 10, and 50 ppbC subsequently for the next four hours. The multipoint calibration period is then ended with two more blank hours to clear the system again. The chromatograms are analyzed by the operator, and in a spreadsheet on the site's computer, the area of the peaks of propane (the calibrant for the PLOT column) and benzene or toluene (depending on the calibrant for the BP column) for each of the concentration hours are entered. The spreadsheet then calculates response factors (RFs) for each concentration and averages them. The percent relative standard deviation between the four response factors must be less than 10% for the multipoint calibration to pass.

2. Carbonyls – ATEC 8000 Cartridge Sampler

   At least annually or following mass flow controller (MFC) calibration verification failure/major repair, the ATEC 8000 MFCs are calibrated against a flow transfer standard. The flow set point for the carbonyl sampling process is 1 liter per minute, and this set point is entered into the ATEC software. The MFC is manually triggered within the software, and a flow verification device that has been calibrated to a NIST-traceable flow transfer standard measures the flow. The difference between the measured flow and the flow displayed on the ATEC must be less than 10% for the calibration to pass. The flow is calibrated by setting the flow on the ATEC until it reads 1.0 liter per minute on the flow verification device, then entering the slope and intercept between the two measurements so that the ATEC reads as close to what the flow verification device shows as possible.

3. True NO_2:

   All gaseous analyzers are calibrated semi-annually to the ideal calibration curve using certified calibration systems with NIST-traceable, NO standards. NO_2 is generated by gas phase titration with excess NO. Four calibration points plus zero are generated with three of the four concentrations in the audit ranges between zero and 80% of the span value (which is the fourth audit value). After performing all necessary maintenance and quality control checks specific to the instrument, the analyzer is zeroed by using zero air from the station calibrator. Then, the highest calibration point is introduced at the beginning of the sample train, including sample line and inlet filter, and the analyzer is adjusted until its output response equals the generated calibration concentration. If the analyzer does not show any drift after about 20 minutes, the point is considered stable. To verify output linearity, the remaining calibration concentrations are introduced in descending order and zero output is re-verified.

Assessment and Oversight

Assessments and Response Actions

Meteorological Audit Frequency and Acceptable Limits

Parameter	Frequency	Acceptable Limits
Mixing height	annually	≤ ± 5 m or ± 1%, whichever is greater
Precipitation	annually	≤ ± 10% of input volume
Relative Humidity	annually	≤ ± 1.5°C (2.7°F) of the dew point temperature
Barometric pressure	annually	< ± 3 hPa
Solar radiation	annually	≤ ± 5% during sunny conditions

Data Management

PAMS AutoGC data is generated separately on an AutoGC-dedicated computer at the remote monitoring site that has the required software necessary for the AutoGC to run (including but not limited to Markes Instrument Control and Agilent OpenLab CDS EZChrom). This collection of software both controls the AutoGC and generates the data from each sample run. On a regular basis throughout the day, the hourly AutoGC data is uploaded to a database specially designed for AutoGC data called ORSAT MAX. On a daily basis, a whole day's worth of AutoGC data is placed

in a ZIP file. These ZIP files are downloaded manually by an air monitoring section employee into the DAQ file server. PAMS Carbonyls data is generated, validated, and entered into AQS by the EPA national contractor (ALS).

True NO_2 will handle in a similar way to the other continuously gaseous analyzers.

Ceilometer data: a python script is run daily through the task manager to get data from .NETCDF and create data files. This script auto null codes error values (-999 is what the instrument reports to indicate an error). A second python script is run daily to send those files to the Linux computer at the Hawthorne station via FTP, then AirVision polls the Linux computer daily to get the mixing layer height (MXLH) data. The ceilometer plots are reviewed weekly to assure that the instrument is working correctly and the months' worth of the data is reviewed in AirVision and graphed along with other meteorological parameters. The MXLH is sorted low to high to identify possible anomalies before the AQS file is created and reported.

Reports to Management

Management is advised of routine operational performance and any maintenance needs. Major issues will be elevated to senior management as necessary.

Data Validation and Usability

Data Review, Verification, and Validation Requirements

The ORSAT MAX program/system provides some automated flagging/alerts for data that fall outside of certain basic parameters (sample timing, CVS failures, etc.). An initial data review is performed by the field operator to ensure that it is being collected successfully and that quality control parameters are met. Independent data validation is performed to determine the necessity of adding qualifying codes prior to uploading to AQS.

Identifying Compromised Data in AQS

Qualifiers associated with PAMS data are still in development by EPA. EPA periodically updates the AQS qualifier list that is published at the following link:

https://aqs.epa.gov/aqsweb/documents/codetables/qualifiers.html

In situations when no qualifier or combination of qualifiers exists to adequately describe the valid compromised data, the data should be estimated by adding the QA qualifier LJ to the data. When invalidating compromised data and no null qualifier exists to describe the rationale for invalidation, the data should be invalidated with the null qualifier AM.

Flagging Data in AQS: Compromised monitoring data will be flagged in AQS only if the data are considered valid for most purposes and uses. AQS permits users to label each data point with up to ten QA qualifiers and/or informational qualifiers.

Invalidating Data in AQS: Data of uncertain origin, data with unacceptable levels of uncertainty, or data which are known to not be an ambient measurement will not have an associated measurement value included in AQS. Such data may be the result of instrument failure, known instrument contamination, irrecoverable data, data corruption, or other issues. If reported to AQS, data generated from routine QC checks (except auto-GC system blanks), calibration, determination of MDLs, or instrument troubleshooting for continuous measurement methods will be coded with a null qualifier to ensure it is not inadvertently reported as ambient data. Invalid data are reported to AQS with a null code qualifier which eliminates the associated measurement parameter and

indicates the reason for the invalidation. AQS accepts a single null qualifier and does not permit addition of other qualifiers to the data point.

Data will be qualified and estimated with descriptive QA and inform flags where the data are compromised but remain valid. In general, such qualification is preferable to invalidation as there remains a measurement value for the data user to access. The data user can then determine whether to use the data value based on the information indicated by the associated qualifier(s). Invalidation removes the measurement entirely from the data point and is therefore of minimal use to an end data user.

Appendix A

References

U.S. Environmental Protection Agency. 2001a. EPA Requirements for QA Project Plans (QA/R-5) (EPA/240/B-01/003).

U.S. Environmental Protection Agency. 2008. Quality Assurance Handbook for Air Pollution Measurement Systems (EPA-454/B-08-002).

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U.S. Environmental Protection Agency. 2017. QA Handbook for Air Pollution Measurement Systems Volume II Ambient Air Quality Monitoring Program (EPA/454/B-17-001).

U.S. Environmental Protection Agency. 2018. Guide to Writing Quality Assurance Project Plans for Ambient Air Monitoring Networks (EPA-454/B-18-006).

U.S. Environmental Protection Agency. 2020. Quality Assurance Project Plan for the Photochemical Assessment Monitoring Stations Required Site Network for Speciated Organic Compounds, Carbonyls, and Meteorological Parameters Including Mixing Layer Height (EPA-454/B-19-003).